Avinash Batchu

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PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years of experience developing enterprise platforms and backend services using Java, Spring Boot, Python, SQL, and REST APIs. Experienced in metadata-driven code generation, API versioning, reusable Maven runtime libraries, database-backed idempotency, and production troubleshooting. Independently developed a modular Python/FastAPI backend integrating Neo4j graph operations and TCS MasterCraft MCCI agent workflows, with structured event processing and Server-Sent Events (SSE). Delivered platform improvements that reduced API-modelling effort and application-generation time by approximately 50% each.

WORK EXPERIENCE

Systems Engineer (Backend Development) 06/2023 – Present

Tata Consultancy Services(TCS)

Product: TCS MasterCraft TransformPlus Forward Engineering Edition (FEE) *Enterprise platform for metadata-driven code generation and Java Spring Boot application development.*

V9–AI-Integrated Backend (Under Development) Python, FastAPI, Neo4j, PostgreSQL, OMgen KDM, Cypher, Server-Sent Events (SSE)

- Independently developed a modular Python/FastAPI backend for the next-generation low-code/no-code platform, providing a middleware layer connecting the UI, databases, and AI-related application capabilities.
- Integrated backend services with TCS MasterCraft MCCI's visual AI-agent platform through exposed APIs, invoking agents, parsing and processing structured streaming events, and delivering processed events to the frontend for real-time AI workflow updates.
- Organized backend functionality into modular router and service layers for graph operations, AI-agent invocation, and file management; implemented Server-Sent Events (SSE) streaming and structured logging across application, request/response, streaming, and database workflows, with time- and size-based log rotation.
- Engineered multi-database Neo4j access with transaction handling and batch Cypher operations using UNWIND to support graph-based workflows across segregated databases.
- Developed a metamodel conversion utility to transfer the V8 metamodel from PostgreSQL to Neo4j, enabling exploration of the existing application metamodel in V9.
- Leveraged Microsoft Copilot in Work mode to assist with V9 Python/FastAPI backend development, using engineering judgment to evaluate and refine AI-assisted implementation work.

V8– Enterprise LCNC Platform Java, Spring Boot, OSC, Eclipse RCP, Kafka, Oracle, PostgreSQL

- Architected and delivered a metadata-driven REST API versioning feature in collaboration with key stakeholders, deriving new API versions from existing metadata models while supporting model/schema changes and preserving backward compatibility; reduced API modelling effort.
- Enhanced Java/Spring Boot code-generation capabilities to automate REST APIs, persistence layers, Oracle/PostgreSQL DDL, Swagger/OpenAPI specifications, and configuration artifacts, reducing application-generation time by approximately 50%.
- Owned OSC scripting for Java code generation, developing and maintaining scripts compiled into JAR artifacts within the platform's code-generation workflow.
- Developed and enhanced reusable Maven-based runtime libraries for generated Java/Spring Boot microservices, including database-backed idempotency using UUID-based request identification, request metadata, and persisted request state/results to determine duplicate-request behaviour.
- Investigated root causes and resolved 150+ production issues across Java code generation, Oracle/PostgreSQL, Eclipse RCP, Kafka, and Swagger/OpenAPI.
- Supported 15+ backend applications by diagnosing application-specific code-generation failures, resolving platform defects, and troubleshooting issues caused by metadata changes.

EDUCATION

Bachelor of Technology in Information Technology 08/2019 – 05/2023
VNR Vignana Jyothi Institute of Engineering and Technology

PROJECTS

Padmaja Palm – Resort Event & Media Management Platform
Django | React | PostgreSQL (Supabase) | Cloudinary | Render | Netlify

- Developed a full-stack resort event and media management platform using Django, React, and PostgreSQL (Supabase), with role-based administration and cloud deployment.
- Implemented staff-scoped media authorization and Cloudinary-signed download URLs with a five-minute expiry to restrict access to protected media.
- Created a storage analytics service integrating Cloudinary APIs to aggregate per-event and account-wide media usage through paginated resource retrieval and quota reporting.
- Engineered media lifecycle management with image/video limits, file-size validation, minimum-image constraints, and cascading cleanup across PostgreSQL and Cloudinary to prevent orphaned resources.

SKILLS

Languages — Java | Python | SQL | Cypher

Backend & Frameworks — Spring Boot | FastAPI | Django | REST APIs | Server-Sent Events (SSE)

Databases — PostgreSQL | Oracle | Neo4j

Platform Engineering — Metadata-Driven Development | Code Generation | API Versioning | Microservices | Maven based runtime libraries | Idempotency

Engineering Fundamentals — Object-Oriented Programming | Design Patterns | System Design | Data Structures and Algorithms | Debugging | Production Troubleshooting

Tools — Git | Maven | Eclipse RCP | Swagger/OpenAPI

ACHIEVEMENTS

- HackerRank certified in Algorithms, Python, SQL, and Java.
- Winner of Blind Code 2020, a national-level coding competition conducted by JNTUH.

LEADERSHIP AND MENTORING

ACM VNRVJIET 2021 – 2022
Treasurer

- Managed operations for a **450+ member student chapter** and organized **15+ technical events**, including workshops and a national-level coding competition.

SMART INTERVIEWS
Teaching Assistant(DSA)

- Mentored **100+ students** in Data Structures and Algorithms, explaining core concepts and guiding students through coding exercises and algorithmic questions.